FEAT-ZD-48600 · DES-02 Design Document · Approved final Figma screens

Design agent · item feat.screens · source hobasa-design-agent/outputs/FEAT-ZD-48600/feat.screens.md · approved by Architect sim.architect, UX lead sim.ux_lead

feat.screens: FEAT-ZD-48600

Final screen specification for FEAT-ZD-48600 (ZD-48600, "Show which overdue invoices belong to vendors that usually pay late"). This is a draft for the DES-02 gate (Architect, UX lead).

Delivery form. "Figma is not connected in this test run" (org.design_library), and no Figma connector is approved for this step (record:brief). A Figma MCP connector was requested in this step (capability request; see feat.design DQ-5). Until it is approved, the final screens are delivered here in text form: the exact HTML each state renders, annotated frames, and the screen rules. When Figma is connected, each frame below maps one to one to a Figma frame with the same name.

This file promotes the approved low-fidelity prototype (feat.prototype WF-1, WF-2) to a build-ready screen specification. It adds no function, data or behaviour beyond feat.prd and feat.trd. Statements marked "design choice (agent judgement)" are for the UX lead to accept or change.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Gate	Architect, UX lead	record:brief
Inputs	feat.prototype (approved), org.design_library, feat.prd, feat.trd	record:brief
UI repository	ap_portal (Node.js 22 CommonJS)	org.architecture; feat.trd (C-8)
Component	`renderOverdueRisk(rows)` in `src/ap_portal/index.js`	feat.trd (IF-7); feat.design (B4)
Screens	One: SC-1 Overdue risk table	feat.prototype (section 2)

1. Screen inventory

ID	Screen / component	Frames	States	User story	Source
SC-1	Overdue risk table in the AP portal	FS-1 Populated, FS-2 Empty	ST-1 to ST-5 from the prototype; ST-6 to ST-8 defensive, new here (section 5)	US-1	feat.prototype (sections 2, 3, 5); feat.prd (US-1, AC-19 to AC-21)

The host page (navigation, title, filters, placement) is not specified in any input and is not designed (feat.prototype PQ-P2). Loading, error, pagination and long-list states are not defined in the PRD and are not designed (feat.prototype PQ-P3; feat.prd PQ-4).

2. Design library applied

The DL-1 to DL-4 numbering is from feat.prototype (section 1); the quoted rules are from org.design_library.

#	Design-library rule (quoted)	Final screen	Source
DL-1	"overdue risk table uses class 'overdue-risk'"	Root element `<table class="overdue-risk">`	org.design_library

#	Design-library rule (quoted)	Final screen	Source
DL-2	"each row has class 'risk-high', 'risk-medium', 'risk-low' or 'risk-unknown'"	Every body row `<tr>` carries exactly one of the four classes; any other `risk` value renders `risk-unknown` (feat.trd IF-7.4)	org.design_library; feat.trd (IF-7.4)
DL-3	"columns Invoice, Vendor, Days overdue, Amount (two decimals)"	Header cells in this order and wording; amount always two decimals, from the CT-1 string	org.design_library; feat.prd (AC-19)
DL-4	"render nothing when there are no rows"	Empty string; no table, header or message	org.design_library; feat.prd (AC-21)

Variables (tokens). The design library defines no colour, spacing or typography variables and no visual style for the `risk-*` classes (org.design_library; feat.prototype PQ-P1). The final screen therefore applies **no tokens and no inline styles**: it emits only the class hooks above, so the portal stylesheet can style them once tokens exist. No colour is invented here.

3. Structure (exact markup)

Design choice carried from feat.trd IF-7.2: no `<thead>`/`<tbody>`, header row without a class, matching the plain style of the existing `renderAgingSummary` (tool:repo, `index.js`). Pending TQ-7.

```
table.overdue-risk
■■■ tr                      (header; no class)
■   ■■■ th  "Invoice"
■   ■■■ th  "Vendor"
■   ■■■ th  "Days overdue"
■   ■■■ th  "Amount"
■■■ tr.risk-{high|medium|low|unknown}   × one per row, in the given order
    ■■■ td  {id, HTML-escaped}
    ■■■ td  {vendor, HTML-escaped}
    ■■■ td  {days, integer text}
    ■■■ td.amount  {amount, two-decimal string}
```

Cell	Content rule	Source
Invoice	`row.id`, HTML-escaped (&, <, >, ", ')	feat.trd (IF-7.3, IF-7.7)
Vendor	`row.vendor`, HTML-escaped	feat.trd (IF-7.3, IF-7.7)
Days overdue	`row.days` as integer text; calendar days after the due date, always more than 30	feat.trd (IF-7.6, IF-6); record:memory (D-2026-031)
Amount	`row.amount` as a string, padded to two decimals as a string if shorter; never `Number()`, `parseFloat` or `formatAmount`; cell class `amount`, as in `renderAgingSummary`	feat.trd (IF-7.3, IF-7.5); record:memory (D-2026-044); tool:repo

4. Frames

FS-1: SC-1 Populated (states ST-1, ST-3, ST-4, ST-5)

Synthetic data from the approved prototype dataset (feat.prototype section 6); the values are illustrative, not facts.

Input (CT-1 rows):

```
[
  {"id": "SYN-INV-004", "vendor": "Synthetic Vendor A", "days": 75, "amount": "1250.00", "risk": "high"},
  {"id": "SYN-INV-002", "vendor": "Synthetic Vendor B", "days": 42, "amount": "980.50", "risk": "medium"},
  {"id": "SYN-INV-003", "vendor": "Synthetic Vendor D", "days": 42, "amount": "15.00", "risk": "unknown"},
  {"id": "SYN-INV-001", "vendor": "Synthetic Vendor C", "days": 31, "amount": "300.00", "risk": "low"}
]
```

Exact output of `renderOverdueRisk(rows)` (one line in the real output; wrapped here for reading):

```
<table class="overdue-risk">
<tr><th>Invoice</th><th>Vendor</th><th>Days overdue</th><th>Amount</th></tr>
<tr class="risk-high"><td>SYN-INV-004</td><td>Synthetic Vendor A</td><td>75</td><td class="amount">1250.00</td></tr>
<tr class="risk-medium"><td>SYN-INV-002</td><td>Synthetic Vendor B</td><td>42</td><td class="amount">980.50</td></tr>
<tr class="risk-unknown"><td>SYN-INV-003</td><td>Synthetic Vendor D</td><td>42</td><td class="amount">15.00</td></tr>
<tr class="risk-low"><td>SYN-INV-001</td><td>Synthetic Vendor C</td><td>31</td><td class="amount">300.00</td></tr>
</table>
```

Annotated frame:

```
 AP portal  (host page not specified: PQ-P2)
                 table.overdue-risk
                ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                ■ Invoice     ■ Vendor                ■ Days overdue ■      Amount ■
                ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
 [risk-high]    ■ SYN-INV-004 ■ Synthetic Vendor A    ■           75 ■     1250.00 ■
 [risk-medium]  ■ SYN-INV-002 ■ Synthetic Vendor B    ■           42 ■      980.50 ■
 [risk-unknown] ■ SYN-INV-003 ■ Synthetic Vendor D    ■           42 ■       15.00 ■
 [risk-low]     ■ SYN-INV-001 ■ Synthetic Vendor C    ■           31 ■      300.00 ■
                ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
 Annotations (designer notes, not rendered):
  - [risk-…] tags are the row class only; there is no visible Risk column or text (AC-19, OQ-B9).
  - Order is set by ap_aging: days descending, then id ascending (SYN-INV-002 before SYN-INV-003).
    The portal keeps the given order and does not re-sort.
  - SYN-INV-003: vendor has no risk entry, so risk-unknown (AC-17).
  - Not listed (filtered in ap_aging): SYN-INV-005, exactly 30 days (AC-14); SYN-INV-006, paid (AC-15).
  - No row actions: no pay, post, approve or chase (NFR-6; PQ-P7).
```

Sources: feat.prototype (WF-1, section 6); feat.prd (AC-14 to AC-17, AC-19, AC-20, NFR-6); feat.trd (IF-5.6, IF-7.2 to IF-7.8).

FS-2: SC-1 Empty (state ST-2)

Input: `[]` (or no rows). Output: `''` (empty string).

```
 AP portal  (host page not specified: PQ-P2)
 ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
 ■   (nothing rendered: renderOverdueRisk returns "")                 ■
 ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
 Annotation: no table, no header, no "no results" message. The bracketed text is a designer note.
```

Sources: feat.prototype (WF-2, ST-2); org.design_library (DL-4); feat.prd (AC-21); feat.trd (IF-7.1).

5. States

ID	State	Trigger	Rendered	Source
ST-1	Populated	`rows` non-empty	FS-1 structure	feat.prototype (ST-1); feat.prd (AC-19)
ST-2	Empty	`rows` empty or missing	`''` (FS-2)	feat.prototype (ST-2); feat.prd (AC-21)
ST-3	Row risk variants	`risk` is `high`, `medium`, `low`, `unknown`	Same row; only the class changes	feat.prototype (ST-3); feat.prd (AC-20)
ST-4	Unknown vendor risk	Vendor not in `risk_by_vendor`	Row class `risk-unknown`	feat.prototype (ST-4); feat.prd (AC-17)
ST-5	Days tie	Equal `days`	Order as received (id ascending from ap_aging)	feat.prototype (ST-5); feat.prd (AC-16, P-A2)
ST-6	Defensive: unexpected `risk`	`risk` outside the four values	Row class `risk-unknown`	feat.trd (IF-7.4, TQ-6)
ST-7	Defensive: markup in text	`id` or `vendor` contains <, &, " and similar	Escaped text, for example `Synthetic Vendor &lt;E&gt;`	feat.trd (IF-7.7, OQ-B8)
ST-8	Defensive: short amount	`amount` `"980.5"`	`980.50`	feat.trd (IF-7.5, TQ-11)

ST-6 to ST-8 are not visible design states; they are rendering safeguards the TRD specifies, listed so the UX lead sees every output the component can produce.

6. Copy

Element	Text	Source
Header 1	Invoice	org.design_library; feat.prd (AC-19)
Header 2	Vendor	org.design_library; feat.prd (AC-19)
Header 3	Days overdue	org.design_library; feat.prd (AC-19)
Header 4	Amount	org.design_library; feat.prd (AC-19)
Empty state	None	org.design_library (DL-4)

Use "overdue"; never the regulatory label "Late" (feat.prd NFR-10; OQ-B2 open).

7. Accessibility and interaction

Item	Final screen	Source
Risk signal	Row class only while OQ-B9 is open. If the portal later styles `risk-*` by colour alone, risk would be conveyed by colour only; the UX lead should decide visible text or a non-colour cue with OQ-B9 / PQ-P4	feat.prd (AC-19, OQ-B9); feat.prototype (PQ-P4)
Header cells	Real `<th>` cells (feat.trd IF-7.2). Agent judgement: this lets assistive technology announce the column headings	feat.trd (IF-7.2)
Interaction	None: no row actions, links or buttons; the risk label is advisory only	feat.prd (NFR-6); feat.prototype (PQ-P7)
Amount alignment	Right-aligned in the prototype (design choice). The markup carries `class="amount"` and no inline style. Design intent (agent judgement): the portal stylesheet, which no input shows, aligns it through that class	feat.prototype (section 7); tool:repo (`renderAgingSummary` uses `class="amount"`)

8. Check against the approved prototype and PRD

Item	Prototype / PRD	Final screen	Result
Table class, row classes, headings, empty	DL-1 to DL-4; AC-19 to AC-21	Sections 2 to 5	Pass
Order and tie	AC-16; ST-5	FS-1 order as given	Pass
Amount without float	NFR-3; prototype "Display the amount string exactly as given"	String display; padding of a one-decimal string is the TRD safeguard (IF-7.5)	Pass, with TQ-11 open on padding
Visible risk	AC-19; OQ-B9	Class only	Follows AC-19; OQ-B9 open
No actions	NFR-6; PQ-P7	None	Pass
Escaping	NFR-7; OQ-B8	Escaped (TRD default)	Follows TRD; OQ-B8 open
Fidelity change	WF-1, WF-2 were low fidelity	Same layout, now with exact markup; no tokens added	No conflict

9. Open questions (non-blocking)

ID	Question	Source
SQ-S1	Approve the Figma connector so FS-1 and FS-2 can be published as Figma frames (requested in this step)	org.design_library; feat.prototype (PQ-P6)

ID	Question	Source
—	Carried: PQ-P1 (tokens for `risk-*`), PQ-P2 (host page), PQ-P3 (other states), PQ-P4 / OQ-B9 (visible risk), PQ-P7 (row actions), OQ-B8 (escaping), OQ-B10 / TQ-11 (amount padding), TQ-6 (class allow-list), TQ-7 (header form), OQ-B2 ("overdue" vs "Late")	feat.prototype (9.1); feat.prd (6.1); feat.trd (7.2)